Do Students Agree with LLM Feedback on Their Own Code? A Within-Subject Study of Four Feedback Configurations in Software Project Courses

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Abstract. LLM-generated feedback can support comprehension checks on students' own software projects, but perceived quality and willingness to use the feedback need not coincide. We evaluated PACE, a repository-grounded question-and-feedback system, with 86 students from 28 team repositories. Each student answered four questions and rated one instance of each of four feedback configurations assigned by random permutation: Sandwich, Rubric, Explainable and Researching. The configurations differ in content structure and, for Researching, the intended model pipeline. Across 344 instances, an exploratory fourteen-item composite averaged 4.07/5 (95% repository-cluster bootstrap CI [3.90, 4.23]); 79.9% of ratings endorsed the AI's judgement, 68.0% endorsed future use and 41.9% endorsed motivation. Configuration differences in the composite were small (Friedman $\chi^2(3) = 23.04$, $p < .001$, Kendall's $W = .089$). The configurations designed to include explicit next steps had higher mean actionability, but the design does not isolate the effect of next steps. All six pairwise 90% confidence intervals for the composite lay within an exploratory ±0.25-point margin; no difference was detected on the Acceptance item. An observational comparison of raw-code and graph contexts was inconclusive ($b = 0.08$, 95% CI $[-0.04, 0.20]$). These findings concern students' perceptions, rather than expert-verified correctness, actual revision or learning gains.

Keywords: LLM-generated feedback · Feedback acceptance · Programming education · Software engineering education · Human–AI trust

1 Introduction

Feedback is one of the strongest levers on learning, but its effect depends on what the learner does with it [11,29]. A comment that is correct yet ignored, or that the student does not believe, changes nothing. Research on feedback literacy therefore treats the student's uptake—making sense of a comment, judging it, and deciding to act—as part of the feedback process rather than its aftermath [1,32].

LLMs can identify issues in student code and explain them in readable language, but their comments may also be misleading [13,18,3]. Guardrails can support guidance without supplying complete solutions [23]. Their growing role in computing education [5,16]

makes student acceptance worth examining separately from technical quality. Students often find AI feedback clear and specific [33,9], yet a Java-course experiment favoured instructor feedback in usefulness and improvement [8], and disclosing AI provenance can alter perceptions [25].

Two gaps motivate this study. First, project courses assess evolving team repositories that no student wrote alone. Questions about learners' own programs can probe comprehension [22], but perceptions of LLM feedback on those answers remain underexplored in the literature discussed here. Second, deployed systems bundle praise, criteria, evidence and model orchestration into different configurations, each with a pedagogical rationale [26,15,17,7]. We compare students' ratings of these bundles without assuming identical underlying judgements.

We address these gaps with PACE (Project Assistance and Course Evaluation), a platform deployed in software project courses at FPT University. PACE analyses each team's repository, generates scenario questions about business rules implemented in it, and returns LLM feedback on every answer. Each student answered four questions and received feedback in four different configurations, assigned by random permutation, then rated every feedback instance on fifteen items covering perceived quality, utility, trust, affect, and acceptance. We ask:

RQ1 How do students judge the quality of LLM feedback on their answers about their own code, and how far do they accept it?
RQ2 Does student agreement differ across the four deployed feedback configurations?
RQ3 How is student agreement associated with raw-code versus graph-based question contexts in the deployed system?

Our contributions are a repository-grounded comprehension workflow, a field evaluation with 86 students and 344 rated instances, and an analysis separating perceived quality, acceptance and motivation. We compare complete configurations, examine response-pattern sensitivity and report uncertainty in an observational context comparison.

2 Related Work

2.1 LLM-Generated Feedback in Programming Education

Hellas et al. found that GPT-3.5 often identified an actual issue in novice programmers' requests but also missed issues or invented them [13]. Kiesler et al. similarly cautioned that formative programming feedback could mislead novices [18]. Dai et al. compared ChatGPT and instructor feedback [3], while CodeHelp demonstrated guarded assistance at course scale [23]. We complement these evaluations with student perceptions of feedback on comprehension answers about team repositories, extending questions about learners' own code [22].

2.2 Student Perceptions and Acceptance of AI Feedback

Usefulness and trust are related to intentions to continue using a system [4,31], while appropriate reliance requires calibration to actual reliability [21]. Algorithm aversion [6]

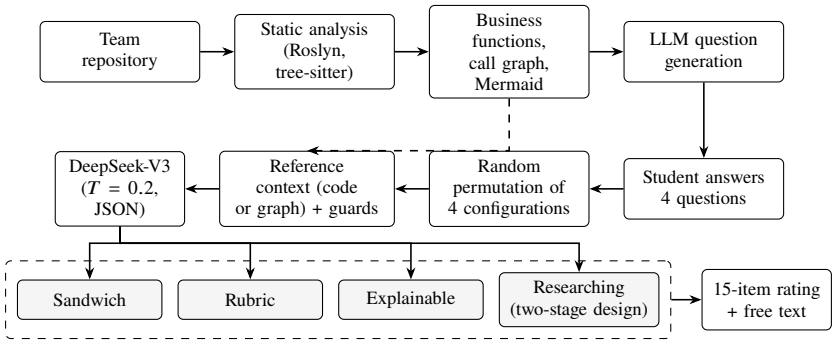


Fig. 1. Overview of PACE. Questions and the evaluator's reference context are both derived from the student's own repository. Each student's four answers receive the four configurations in a random order.

and appreciation [24] both occur. In programming education, Zhang et al. reported preferences for code-referencing feedback [33]; Er et al. found advantages for instructor feedback [8]; and Nazaretsky et al. found perceptions sensitive to AI provenance [25]. These findings motivate measuring trust and intention alongside perceived quality.

2.3 Feedback Configurations

Sandwich feedback improved perceptions without improving performance in a controlled study [26]. Rubrics make criteria explicit [15]. Evidence-based explanations may support appropriate trust [17]. Multiagent debate has improved factuality [7], while model-as-judge evaluations have shown agreement with human preferences [34]. These results motivate our configurations but do not establish their effectiveness in this setting. Our within-student comparison evaluates complete bundles.

3 The PACE System

Figure 1 summarises the pipeline. PACE consists of an ASP.NET Core (.NET 9) back end, a PostgreSQL database and a Next.js front end; the parts relevant to this study are repository analysis, question generation, and feedback generation.

3.1 Repository Analysis and Question Generation

For each team repository, PACE clones the submitted branch and parses it with Roslyn (C#) or tree-sitter (Java, JavaScript and TypeScript). Endpoints are grouped into *business functions* (e.g., "contract management" or "payment confirmation"), either from a template stored in the repository or from a draft that an instructor reviews and confirms. For every function, PACE resolves calls from each entry point and stores both the source code of the involved methods and a data-flow graph with ordered call edges, branch conditions and a Mermaid rendering.

Questions are generated per business function by the configured question-generation LLM (temperature 0.3, structured JSON output), which receives either the raw source code or the Mermaid graph; a router selects the graph when the flow involves more than ten distinct methods and the raw code otherwise. The prompt targets scenario questions about guard conditions and failure branches, for example: *"When the facility manager tries to swap the storage unit of a contract that still has an outstanding balance, how does the system respond and why is the action blocked?"* Each question is stored with a suggested answer and with the context type used to produce it.

3.2 Feedback Generation

When a student submits answers, each answer is evaluated against a *reference context* built from the same business function: the raw source of its methods, or its flow graph with numbered steps and conditions, following the context type of the question. The system prompt fixes an instruction hierarchy: content inside the `<student_answer>` tag is data, never instructions; the reference context and the suggested answer supply the evaluation reference, whose correctness is not independently established here; the model comments and explains but does not assign an overall grade; and the output must be a single JSON object. An input guardrail rejects answers that contain prompt-injection phrases or executable payloads, and an output guardrail validates the JSON before it is stored.

The four configurations bundle different content requirements and generation procedures:

- **Sandwich**: positives, constructive criticism, and next steps.
- **Rubric**: a list of criteria that the model names for the question (e.g., understanding of the business flow, identification of the guard condition), each with a 1–5 score and a comment.
- **Explainable**: a *code evidence* field that cites methods, steps or conditions from the context, and a *logic explanation* that relates them to the answer.
- **Researching**: a configuration designed as a two-stage ensemble in which DeepSeek-V3 first inspects the answer against the code and GPT-OSS-120B (served by Groq) then writes a combined report: an overall comment, positives, criticism, next steps, code evidence, two scored criteria, and a consensus summary with pedagogical guidance.

Only Sandwich and Researching are designed to require an explicit list of next steps. A schema audit found one Rubric-labelled payload with next-step fields but no criterion list; all analyses retain its assigned label. The other 85 Rubric payloads contained criteria. The three single-model configurations use DeepSeek-V3 (`deepseek-chat`; temperature 0.2, JSON mode). If the second model in the Researching pipeline fails, the system falls back to the single-model path; all 86 Researching records list both model roles in the nested `researching.models_involved` metadata. This supports the recorded two-model configuration; independent provider execution logs are not part of the export.

The assigned configuration is the experimental manipulation. Recorded model roles describe provenance metadata rather than an independent execution audit. When a student submits $n = 4$ answers, PACE shuffles the four configurations and assigns one to each answer, so every student sees every configuration exactly once, in random order.

Table 1. Survey instrument (translated from Vietnamese). †Reverse-worded.

Construct	Item	Statement
Perceived quality	Accuracy	The AI's comment correctly identifies the problems in my answer.
	Specificity	The AI points to exactly which part of my answer needs revisiting, rather than commenting in general.
	Clarity	The wording and presentation of the comment are clear and unambiguous.
	Transparency	I understand which academic criteria the AI used to assess my answer.
Perceived utility	Usefulness	This AI tool is an effective learning assistant for my project work.
	Actionability	I know exactly which technical steps to take to improve my answer.
Trust	Competence	I trust the technical expertise and analytical capability of this AI.
	Reliability	The AI assesses consistently and does not change its standards at random.
	Fairness	The AI's assessment fairly reflects the effort and actual quality of my answer.
Affect	Affect	I feel motivated to keep learning after reading the AI's comments.
	Psych. safety	AI feedback causes less pressure than being criticised by a person.
Acceptance & intention	Acceptance	I agree with the AI's judgement of my answer's strengths and weaknesses.
	Willingness	Based on this comment, I am ready to revise my answer right away.
	Behav. intention	I intend to keep using this LLM system for feedback in future projects.
Separate item	Rejection†	I tend to want to refute or argue against the AI's assessment.

4 Study Design

4.1 Context and Participants

The study took place in October 2026 in undergraduate software engineering project courses at FPT University, Vietnam. The research team visited classes and invited students to use PACE and provide candid feedback as preparation for questions they might encounter before a project defence panel. Students used the system to rehearse explaining their team's repository. Recruitment was classroom-based rather than a probability sample. The sample comprises 86 students from 28 team repositories, answering 190 distinct questions on 152 business functions. Answers were written in Vietnamese and were short, about 51 whitespace-separated tokens on average. Submissions were made between 20 and 26 October 2026.

4.2 Instrument

After reading each feedback instance, students rated it on fifteen statements on a five-point scale (1 = strongly disagree, 5 = strongly agree) and could add a free-text reply. The items were written in Vietnamese by the research team, drawing on formative-feedback principles [29] and technology-acceptance constructs [4,31,21]. Table 1 lists them with English translations, grouped into five a priori constructs. REJECTION is reverse-worded.

4.3 Data Set

The survey export contained 344 answer records from 86 submissions, one submission per student. We screened every record on metadata only (no explicit development-fixture marker in the export, feedback stored, all fifteen ratings present); all 344 passed, so none was excluded. Every student rated all four configurations, which gives 86 complete

blocks and 86 instances per configuration. The export also stores the context type of each question: 280 instances (157 questions) came from graph-based questions and 64 instances (33 questions, 13 repositories) from raw-code questions. Thirty-two students received at least one raw-code question, and 26 of them answered questions of both types.

4.4 Analysis

Scores. The *AI agreement index* (AAI) is an exploratory composite: the mean of fourteen non-reversed items across several domains. It is not a validated unidimensional measure of agreement. Construct scores are item means, and the Acceptance item is also reported separately. The decision to exclude Rejection followed inspection of its inconsistent behaviour (Section 5.1); it is therefore a data-informed scoring choice [30].

Reliability and data quality. We report Cronbach's α [2] and McDonald's ω_t from a one-factor solution [12], and flag *straight-lined* responses—identical ratings on all fifteen items—following Kim et al. [19].

RQ1. We report item means with 95% confidence intervals from a cluster bootstrap that resamples students (5,000 replicates), since each student rated four feedback instances. To test whether agreement exceeds the neutral midpoint, we aggregate to one mean per student and apply a one-sample Wilcoxon signed-rank test against 3, with the matched-pairs rank-biserial correlation r_{rb} as effect size and Holm correction across the fifteen items [14]. To explore conditional associations with acceptance, we fit student fixed-effects regressions after demeaning predictors and outcomes within each student:

$$y_{ij} - \bar{y}_j = \sum_{k=1}^{5} \beta_k (z_{kij} - \bar{z}_{kj}) + e_{ij}, \tag{1}$$

where y is Acceptance, Willingness or Behavioural intention. Predictors are perceived quality, perceived utility, trust, motivation and psychological safety, standardised using their full-sample SDs before demeaning. We separate the two affect items because their joint reliability is low. Percentile 95% confidence intervals resample all 28 repositories with replacement (5,000 replicates), preserving the students and shared questions within each repository. These models describe contemporaneous within-student associations, not causal drivers or differences between students.

RQ2. The primary test is a Friedman test [10] on AAI over the 86 complete blocks, with Kendall's W as effect size, followed by Holm-corrected Friedman tests on each construct and item and Wilcoxon signed-rank tests for the six configuration pairs. Because the Friedman test uses only the within-student rank order, we also fit a linear mixed model, AAI $\sim$ configuration + (1 | student), compared to an intercept-only model by a likelihood-ratio test, and a second model that adds feedback length and answer length as covariates. Because significance says nothing about whether a difference matters, we also run equivalence tests [20]: a pair of configurations meets the exploratory equivalence criterion when the 90% confidence interval of the paired mean difference lies

within ±0.25 points, a quarter of one scale step. This pragmatic, exploratory margin has not been calibrated against learning outcomes or elicited from stakeholders; equivalence claims are restricted to AAI under this margin. A sensitivity analysis repeats the Friedman test without straight-lined responses.

RQ3. Context type was not randomised: the router uses graph context for functions involving more than ten methods and raw code otherwise. Complexity, question context and evaluator context are therefore coupled. We report descriptive means and an exploratory model, AAI ∼ context + configuration + (1 | student). Its coefficient is an adjusted association, not an isolated context effect or a test of equivalence. Row-level Mann–Whitney tests are not used for inference because observations within students are dependent.

Dependence and scope. Answers are assigned within students, who share repositories and questions. Repository resampling preserves students and shared questions: each belongs to one repository. Item intervals, rank tests and student-intercept models do not fully capture repository dependence. Straight-lining is a response-pattern flag, not proof of inattention.

Analyses use Python and statsmodels [28]. Supplementary scripts accompany the paper. Identifier-bearing raw exports are withheld; an anonymised release remains to be prepared.

5 Results

5.1 Reliability and Data Quality

The fourteen non-reversed items had high internal consistency ($\alpha = .936$, $\omega_t = .939$). The subscales were acceptable to excellent (perceived quality $\alpha = .91$, perceived utility .78, trust .80, acceptance & intention .78), with the exception of affect (.56): its two items, motivation and psychological safety, should not be assumed to form a single reliable construct. Internal consistency alone does not establish dimensionality or construct validity. Two observations qualify this.

First, 64 of the 344 responses (18.6%) were straight-lined, and 11 of the 86 students (12.8%) straight-lined all four of their ratings. Excluding straight-lined response rows lowers α only to .90, so the scale's consistency does not rest on straight-lining.

Second, straight-lining hid most of the signal in the reverse-worded Rejection item. Across all responses, Rejection correlated only weakly and negatively with Acceptance (Spearman $\rho = -.14$), and its item–rest correlation was close to zero (−.03). Once straight-lined responses were removed, the negative association was larger ($\rho = -.40$ with Acceptance and $\rho = -.41$ with the agreement index, descriptively). A student who rates every statement 4 also "agrees" to wanting to argue with the AI, a known weakness of reverse wording [30]; we therefore keep the item out of the exploratory composite. These correlations describe response patterns; row-level significance tests would ignore repeated ratings.

Table 2. Item ratings across 344 feedback instances. CI: 95% cluster-bootstrap interval of the mean. Agree/Disagree: % of ratings ≥ 4 / ≤ 2. r_{rb} and p_{Holm}: one-sample Wilcoxon test of student means against the midpoint 3 ($n = 86$). †Reverse-worded, not reverse-scored.

Item	Mean (SD)	95% CI	Agree	Disagree	r_{rb}	p_{Holm}
Accuracy	4.38 (0.96)	[4.21, 4.53]	83.4	5.5	0.98	<.001
Specificity	4.36 (0.94)	[4.18, 4.53]	83.4	5.2	0.96	<.001
Clarity	4.33 (0.88)	[4.18, 4.49]	81.7	3.5	0.99	<.001
Transparency	4.29 (0.95)	[4.12, 4.46]	79.9	4.9	0.97	<.001
Usefulness	4.09 (0.77)	[3.96, 4.22]	84.0	2.9	0.96	<.001
Actionability	4.13 (0.88)	[3.97, 4.28]	80.8	4.9	0.95	<.001
Competence trust	4.17 (0.98)	[4.00, 4.34]	76.5	4.9	0.95	<.001
Reliability	3.88 (0.80)	[3.73, 4.01]	78.5	4.9	0.90	<.001
Fairness	4.15 (0.79)	[4.01, 4.28]	81.4	2.3	0.99	<.001
Affect	3.53 (0.87)	[3.39, 3.68]	41.9	4.9	0.93	<.001
Psych. safety	3.83 (0.85)	[3.69, 3.97]	76.2	7.3	0.90	<.001
Acceptance	4.09 (0.82)	[3.95, 4.22]	79.9	3.5	0.98	<.001
Rejection†	2.87 (1.22)	[2.64, 3.11]	30.8	48.8	-0.21	.110
Willingness	3.97 (0.69)	[3.85, 4.07]	81.4	2.3	0.98	<.001
Behav. intention	3.80 (0.85)	[3.64, 3.94]	68.0	5.2	0.90	<.001

5.2 RQ1: How Students Judge the Feedback

Table 2 and Fig. 2 show that ratings were positive on every non-reversed item. The agreement index averaged 4.07 (SD 0.64, 95% repository-bootstrap CI [3.90, 4.23]) and was above the midpoint for nearly all students ($r_{rb} = .98$, $p < .001$). Students most often endorsed usefulness (84.0% agree), accuracy and specificity (83.4% each), and explicit disagreement stayed at or below 7.3% on all fourteen items. On the reverse-worded item, 48.8% rejected the idea of arguing with the AI and 30.8% endorsed it.

The profile is far from flat. Students readily said that a comment was correct, specific and clear, and most also extended trust to the system's competence (76.5%) and consistency (78.5%). They were less ready to commit to it: 68.0% intended to keep using PACE in future projects. The weakest item by a wide margin was affect: only 41.9% said the feedback made them motivated to keep learning, and more than half (53.2%) chose the neutral option. Endorsement of perceived quality was more frequent than endorsement of motivation.

Conditional associations with acceptance. Table 3 reports within-student coefficients and repository-bootstrap intervals. Utility was positively associated with acceptance and future-use intention. Trust and perceived quality were positively associated with acceptance, but their future-use intervals included zero. Motivation was positively associated with willingness; psychological-safety intervals included zero throughout. The fifteen exploratory intervals are unadjusted, and coefficient differences were not directly tested.

AI criterion scores and student ratings. Among 171 instances with criterion scores (Rubric and Researching), descriptive Spearman correlations were $-.15$ with acceptance, $-.12$ with fairness and $-.07$ with AAI. These descriptive associations mix repeated students and scoring criteria; they do not establish self-serving responses.

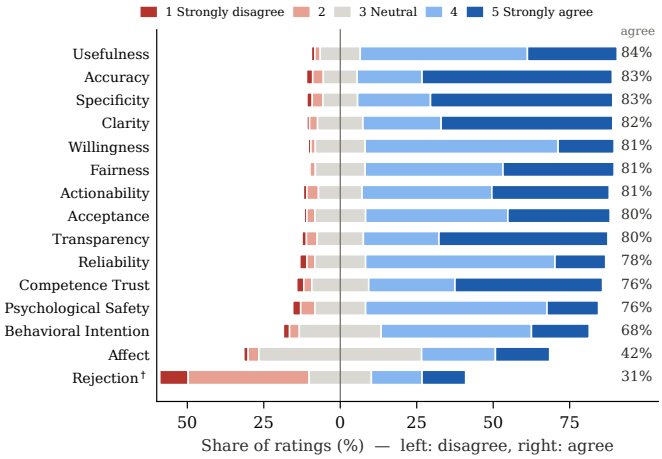


Fig. 2. Distribution of ratings per item, sorted by the share of agreement (right-hand labels). Neutral ratings are centred on zero.

Table 3. Student fixed-effects regressions (Eq. 1). Cells show rating points per full-sample predictor SD [95% repository-bootstrap CI]; 5,000 replicates. Motivation and psychological safety are separate predictors.

Predictor	Acceptance	Willingness	Behav. intention
Perceived quality	0.22 [0.02, 0.44]	0.09 [−0.19, 0.37]	0.07 [−0.15, 0.33]
Perceived utility	0.32 [0.17, 0.46]	0.07 [−0.11, 0.19]	0.23 [0.07, 0.37]
Trust	0.23 [0.09, 0.40]	0.04 [−0.10, 0.21]	0.15 [−0.10, 0.40]
Motivation	−0.10 [−0.21, 0.02]	0.25 [0.01, 0.46]	−0.05 [−0.18, 0.09]
Psych. safety	−0.04 [−0.12, 0.01]	−0.01 [−0.16, 0.08]	0.07 [−0.09, 0.25]

5.3 RQ2: Differences Between Feedback Configurations

The four configurations received similar but not identical ratings (Table 4, Fig. 3). Mean agreement ranged from 4.01 (Rubric) to 4.12 (Sandwich). The Friedman test detected a difference ($\chi^2(3) = 23.04$, $p < .001$), but its size was small ($W = .09$). Pairwise tests located it between Sandwich and the two configurations not designed to require next steps: Sandwich was rated 0.11 points above Rubric ($p_{\text{Holm}} = .002$) and 0.06 points above Explainable ($p_{\text{Holm}} = .023$); no other pair differed after correction. The mixed model, which weighs the size of each difference rather than its rank, was less conclusive (likelihood-ratio $\chi^2(3) = 6.65$, $p = .084$; Rubric vs. Sandwich $b = -0.11$, $p = .013$). Many students rated the configurations almost alike but in a consistent order, which a rank test detects more easily.

The differences were concentrated in two constructs. Perceived utility varied most ($\chi^2(3) = 45.57$, $p_{\text{Holm}} < .001$, $W = .18$): Sandwich (4.27) and Researching (4.22) were rated more useful and actionable than Rubric (4.01) and Explainable (3.95). At the item level this gap was largest for actionability (4.33 and 4.28 vs. 3.97 and 3.95; $p_{\text{Holm}} < .001$), which is consistent with their explicit lists of next steps, although this feature

Table 4. Ratings by feedback configuration, mean (SD). Friedman tests over the 86 students, each of whom rated all four configurations; p for the construct rows is Holm-adjusted. Each configuration has 86 instances.

	Sandwich	Rubric	Explainable	Researching	$\chi^2(3)$	p	W
Agreement index (AAI)	4.12 (0.69)	4.01 (0.66)	4.07 (0.61)	4.09 (0.60)	23.04	<.001	0.089
Perceived quality	4.36 (0.82)	4.33 (0.82)	4.39 (0.79)	4.28 (0.86)	6.31	.195	0.024
Perceived utility	4.27 (0.86)	4.01 (0.69)	3.95 (0.66)	4.22 (0.73)	45.57	<.001	0.177
Trust	4.12 (0.78)	3.91 (0.75)	4.11 (0.70)	4.12 (0.65)	24.98	<.001	0.097
Affect	3.74 (0.69)	3.65 (0.75)	3.63 (0.76)	3.72 (0.68)	10.22	.050	0.040
Acceptance & intention	3.95 (0.70)	3.91 (0.71)	3.96 (0.62)	3.98 (0.62)	3.70	.295	0.014
Feedback length (words)	248	267	215	623	172.19	<.001	0.667

was not manipulated independently. Trust also varied ($\chi^2(3) = 24.98$, $p_{\text{Holm}} < .001$, $W = .10$), with Rubric lowest (3.91 vs. 4.11–4.12); Rubric feedback also received the lowest ratings for competence trust (3.97) and fairness (3.95), and fairness asks whether the assessment reflects the answer's effort and quality. Sandwich feedback was rated highest on affect (3.69 vs. 3.41 for Explainable; item $p_{\text{Holm}} = .016$). No difference was detected for perceived quality or acceptance & intention, and no difference was detected on the ACCEPTANCE item across configurations (4.05–4.12, $p_{\text{Holm}} = 1.00$). Researching feedback had a lower mean clarity rating than the others (4.19 vs. 4.36–4.41), but this did not survive correction ($p_{\text{Holm}} = .22$).

Researching feedback was longer: its mean length was 623 words versus 215–267 for the other configurations ($\chi^2(3) = 172.2$, $p < .001$, $W = .67$). Adding feedback length and answer length to the mixed model left the estimated Rubric–Sandwich gap near -0.11 points ($p = .018$). The covariates were not statistically distinguishable from zero (feedback length $b = -0.06$, $p = .16$; answer length $b = 0.01$, $p = .78$). Feedback length is generated by the assigned configuration, so this adjustment does not identify a direct configuration effect or rule out a role for length. Token and monetary cost comparisons are omitted pending an auditable usage-log extraction.

For all six configuration pairs, the 90% confidence interval of the paired difference in AAI fell inside ± 0.25 points; the most extreme bound was 0.21 (Sandwich vs. Rubric, interval $[0.02, 0.21]$). The Friedman difference remained after excluding straight-lined responses (280 retained responses, of which 64 students supplied complete blocks; $\chi^2(3) = 23.51, p < .001, W = .12$), with the same ordering. A repository-cluster bootstrap also placed all six 90% intervals inside the exploratory ± 0.25 margin; the Sandwich–Rubric interval was $[0.008, 0.212]$. This sensitivity result concerns AAI only and is not an item-level equivalence claim. Excluding the student with a Rubric schema deviation left 85 complete blocks and a similar Friedman statistic ($\chi^2(3) = 23.22$); this post hoc check does not replace the assigned-label analysis.

In the intercept-only model, most AAI variance lay between students: the intraclass correlation of the agreement index was .77 (also .77 without straight-lined responses), indicating substantial clustering by student. This variance decomposition does not identify why configuration differences are small.

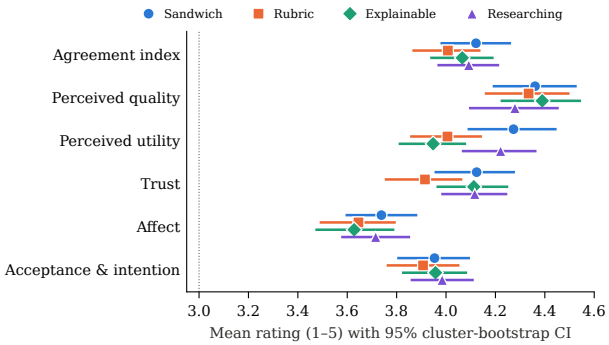


Fig. 3. Agreement index and construct scores by feedback configuration, with 95% cluster-bootstrap confidence intervals. The dotted line marks the neutral midpoint.

Table 5. Agreement by question context type, mean. b: mixed-model estimate for raw code vs. graph, controlling for format, with a random intercept per student.

Context	n	AAI	Quality	Utility	Trust	Acceptance & int.
Graph	280	4.06	4.30	4.11	4.05	3.94
Raw code	64	4.14	4.52	4.12	4.16	4.00
$b_{\text{raw}} = 0.08$ ($SE = 0.06$), $p = .22$; exploratory student-intercept model						

5.4 RQ3: Raw-Code versus Graph-Based Questions

Mean AAI was 4.14 for raw-code and 4.06 for graph-based questions (Table 5). The exploratory model estimated $b = 0.08$ ($SE = 0.06$, $p = .22$, 95% CI $[-0.04, 0.20]$). Among the 26 students exposed to both types, pooled means were 3.99 and 4.01, respectively; these are descriptive rather than a balanced paired estimate. Perceived quality averaged 4.52 versus 4.30, and transparency 4.59 versus 4.22. The routing rule ties context to method count, and the same context type feeds the evaluator. These estimates cannot separate complexity, question generation and feedback context; the non-significant coefficient does not establish equivalence.

5.5 Students' Free-Text Replies

The existing export includes 160 optional free-text replies (46.5% of instances). We use these as illustrative observations rather than a validated thematic prevalence analysis; no independently coded category counts are reported. Replies include proposed revisions and follow-up questions about error codes or the order of checks. One student disputed an error name and stated that validation returns `409 SCHEDULE_CONFIRMED` before the lock check; another objected to the type of request assumed in the feedback. These are students' reported objections, not independently verified error classifications. A further reply described a single block of feedback text as difficult to read. Observed revision and an expert audit are needed to establish whether these comments correspond to uptake or technical errors.

6 Discussion

Acceptance, motivation and reported use. Endorsement of accuracy and usefulness exceeded endorsement of motivation and future use. Utility was positively associated with acceptance and intention within students; trust was associated with acceptance, consistent with technology-acceptance research [4,31]. Motivation was associated with willingness when separated from psychological safety. These self-reports do not establish actual revision or correctness. Specific objections in replies illustrate the need to verify appropriate reliance independently [21].

Configuration bundles motivate controlled tests. Small AAI differences coexisted with an actionability gap of about 0.38 points between Sandwich and Explainable. The configurations designed to include next steps had higher mean actionability, but content, scoring, length and model orchestration also differed. An experiment manipulating next steps independently is needed. Researching produced more text without a detected AAI advantage over Sandwich; efficiency claims require usage logs. The high student ICC supports accounting for individual differences in comparisons.

Response patterns and measurement. Each student supplied sixty ratings; 18.6% of response sets were straight-lined, complicating the reverse-worded item [19]. A shorter instrument requires validation. Attention checks, behavioural probes and provenance markers merit evaluation [30]; straight-lining alone does not establish inattention.

Threats to validity. Internal validity. Assignment within students limits stable individual differences, but each configuration evaluates a different answer. Question difficulty, order and carryover remain possible sources of variation; the comparison cannot isolate individual content features. Exported Researching model roles are not independently corroborated by provider logs. *Construct validity.* The study-specific instrument lacks established factorial validity. AAI spans heterogeneous domains, exclusion of Rejection was data-informed, and the affect pair has low reliability. Straight-lining does not establish inattention. Self-reports collected together may share common-method variance [27]. No observed revision, learning outcome or expert correctness measure was collected, and free-text observations are illustrative rather than validated thematic prevalence estimates. *External validity.* Classroom recruitment at one university, preparation for a defence panel, and novelty may shape responses. RQ3 is observational and confounded by method count and the evaluator's context. *Statistical conclusion validity.* Student blocks are used in primary comparisons, and repository resampling preserves shared-student and shared-question dependence for the added estimates. Rank tests and student-intercept models do not model all of this dependence. Predictor regressions isolate within-student variation, but do not establish causal effects or adjust for every possible time-varying confounder. Mixed models assume interval scaling and serve as sensitivity analyses; the AAI omnibus Friedman and mixed-model tests differ at the .05 threshold. The ± 0.25 equivalence margin is pragmatic, exploratory and specific to AAI. The fifteen exploratory regression intervals are unadjusted, and only 28 repository clusters are available.

7 Conclusion

PACE connects repository analysis, scenario questions and structured feedback in a classroom comprehension workflow. Across 344 instances from 86 students, perceived feedback ratings were positive, but endorsement of future use and motivation was lower than endorsement of accuracy and usefulness. Small AAI differences across four assigned configurations coexisted with larger actionability differences. The bundled configurations do not isolate next steps or model effects, and the context comparison was inconclusive. Findings concern perceptions rather than correctness or learning.

Future work should audit execution logs, validate the instrument, measure revisions and expert correctness, and manipulate next steps and context independently.

Acknowledgments. We thank the students and instructors of the participating software project courses.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

Research ethics and data governance Author confirmation required before submission: classroom recruitment is described in Section 4. Voluntariness, any relationship to grading, consent for research use, institutional review or exemption, and safeguards for student identifiers and repository content remain to be documented. No approval or consent procedure is asserted without supporting records.

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